C73-43: contemporaneously nested information does not ensure approximate stopping equilibrium

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Abstract

We construct a bounded two-player stopping game with dates $0, 1$ on a standard Borel common-prior probability space. Its observation filtrations are contemporaneously nested, but it has no measurable randomized ex-ante 2^{-42}-Nash equilibrium. The counterexample includes every simultaneous stopper set and the no-stop outcome. The proof is self-contained: a quantitative obstruction to a measurable noisy-majority recursion yields a binary Bayesian game without approximate equilibrium; a hidden nature flag creates a common indifference row; and that row permits an exact realization as a stopping game.

1 The target and the result

Problem C73-43 asks whether every bounded finite-horizon stopping game on an arbitrary common-prior probability space, with

$$\mathcal{F}_{n,t} \subseteq \cdots \subseteq \mathcal{F}_{1,t} \qquad (0 \le t \le T),$$

admits an ex-ante ε-Nash equilibrium for every $\varepsilon > 0$. Each player has independent private uniform seeds, independent of the underlying state. A strategy is a randomized stopping time in $\{0, \dots, T, \infty\}$. The first stopping date and the full simultaneous stopper set determine a bounded measurable payoff vector; a separate bounded vector is paid if nobody stops. Public survival is observed. The payoff variables in the displayed problem are arbitrary bounded measurable state functions; no measurability with respect to an individual player's private filtration is imposed. One-step revelation of the richer information to the less-informed player is also not imposed.

Theorem 1. *There is a game in this family with two players and $T = 1$ which has no measurable randomized ex-ante 2^{-42}-Nash equilibrium. Its common-prior space is standard Borel, and its payoff bounds can be chosen as $3/2$ for Player 1 and 1 for Player 2. Consequently, the universal existence assertion in C73-43 is false.*

The contemporaneously nested stopping question was posed in Section 4 of the 2024 version of Jacobovic–Solan [1]. Simon–Tomkowicz's work on Bayesian nonexistence [2] supplies relevant context, but its nonexistence theorem is not a premise of the argument here.

In the direct type encoding used here, the joint signal prior is not dominated by the product of the marginal signal laws. On phase R, of probability $1/2$, Player 1's tree signal is exactly the tree coordinate of Player 2's signal. Under the product of the marginal laws, those two tree coordinates are independent with the nonatomic law m, so equality has probability zero. The non-atomicity follows already from any one uniform tree coordinate. This observation about the direct type encoding makes no assertion about alternative primitive-coordinate representations.

2 A quantitative obstruction on a labeled tree

Let $W = \{1,2,3\}^{<\mathbb{N}}$ be the countable set of finite words, including the empty word e. Define

$$X = (\{0,1\} \times [0,1]^3)^W.$$

At vertex w, write its four coordinates as $(b_w, t_{w,1}, t_{w,2}, t_{w,3})$. Equip X with its countable product Borel sigma field and the product probability m, with each b_w a fair bit and every $t_{w,i}$ uniform on $[0,1]$, all independent. This is a standard Borel probability space. For $i \in \{1,2,3\}$, define the subtree map T_i by

$$b_w(T_i x) = b_{iw}(x), \qquad t_{w,j}(T_i x) = t_{iw,j}(x).$$

The three random trees T_1x, T_2x, T_3x are independent with law m, and are independent of all root coordinates. Set $s(x) = (-1)^{b_e(x)}$. For three signs,

$$\mathrm{Maj}(z_1,z_2,z_3) = \mathrm{sign}(z_1+z_2+z_3) = \frac{z_1+z_2+z_3-z_1z_2z_3}{2}, \qquad z_i \in \{-1,1\}.$$

Lemma 2. *There is no measurable $z : X \to \{-1,1\}$ for which*

$$\delta := m\{z(x) \neq s(x)\,\mathrm{Maj}(z(T_1x), z(T_2x), z(T_3x))\} \leq \frac{1}{4096}.$$

Proof. Put

$$z^*(x) = s(x)\,\mathrm{Maj}(z(T_1x), z(T_2x), z(T_3x)), \qquad \mu = \mathbb{E}_m z.$$

The root sign is fair and independent of the child trees, so $\mathbb{E}_m z^* = 0$. Since z, z^* are signs,

$$|\mu| \leq \mathbb{E}_m|z - z^*| = 2\delta, \qquad \|z - z^*\|_2 = 2\sqrt{\delta}. \tag{1}$$

Let $\mathcal{H}_n$ be generated by the coordinates at vertices of length less than n; $\mathcal{H}_0$ is trivial. Put

$$a_n = \mathbb{E}_m[z \mid \mathcal{H}_n], \qquad v_n = \mathbb{E}_m a_n^2, \qquad a_{n,i}(x) = a_n(T_i x).$$

Conditionally on $\mathcal{H}_{n+1}$, the unexplored portions of the three child trees are independent. Thus

$$\mathbb{E}_m[z^* \mid \mathcal{H}_{n+1}] = \frac{s(x)}{2}(a_{n,1} + a_{n,2} + a_{n,3} - a_{n,1}a_{n,2}a_{n,3}). \tag{2}$$

The variables $a_{n,1}, a_{n,2}, a_{n,3}$ are independent and identically distributed, each with mean μ and second moment v_n. Expanding the square in (2) gives

$$\begin{aligned} \|\mathbb{E}_m[z^* \mid \mathcal{H}_{n+1}]\|_2^2 &= \frac{1}{4}(3v_n + 6\mu^2 - 6v_n\mu^2 + v_n^3) \\ &= \frac{3v_n + v_n^3}{4} + \frac{3}{2}\mu^2(1-v_n). \end{aligned} \tag{3}$$

Conditional expectation contracts the L^2 norm. The triangle inequality and (1)–(3) imply

$$\sqrt{v_{n+1}} \leq \sqrt{\frac{3v_n + v_n^3}{4} + \frac{3}{2}\mu^2(1-v_n)} + 2\sqrt{\delta}. \tag{4}$$

We prove $v_n \leq 1/4$ for every n. Initially, $v_0 = \mu^2 \leq 4\delta^2 < 1/4$. If $v_n \leq 1/4$, then

$$\frac{3v_n + v_n^3}{4} + \frac{3}{2}\mu^2(1-v_n) \leq \frac{49}{256} + 6\delta^2 < \left(\frac{15}{32}\right)^2, \qquad 2\sqrt{\delta} \leq \frac{1}{32}.$$

Hence (4) gives $\sqrt{v_{n+1}} < 1/2$, completing the induction.

The sigma field generated by all $\mathcal{H}_n$ is the full product sigma field. Finite-coordinate functions are dense in $L^2(m)$, and conditional expectations fix each such function for all sufficiently large n. Since conditional expectation is a contraction, this density implies $a_n \to z$ in L^2. Therefore $v_n \to \mathbb{E}_m z^2 = 1$, contradicting $v_n \leq 1/4$. □

3 A two-player binary Bayesian game

Let the state space Ω_0 be the disjoint union

$$G = X \times \{1,2,3\}, \qquad R = X \times \{1,2,3\} \times [0,1].$$

Give phases G, R probability $1/2$ each. Conditionally on G, (x,i) has law m times the uniform law on $\{1,2,3\}$. Conditionally on R, (y,i,t) has the corresponding product law with Lebesgue measure for t.

Both players have actions $\{0,1\}$. Player 1 observes x in state (G,x,i), and y in state (R,y,i,t). Player 2 observes

$$(i, T_i x, t_{e,i}(x)) \quad \text{in phase } G, \qquad (i,y,t) \quad \text{in phase } R. \tag{5}$$

The two conditional laws of Player 2's signal agree: each is the uniform law on i, times m, times Lebesgue measure. Player 1's two conditional signal laws also agree. Consequently, conditional on either player's signal, the two phases each have probability $1/2$.

The state-dependent payoffs are

state	$u_1(\omega, a_1, a_2)$	$u_2(\omega, a_1, a_2)$
(G,x,i)	$a_1 s(x)(3a_2 - 3/2)$	0
(R,y,i,t)	0	$a_2(a_1 - t)$.

(6)

These functions are bounded and Borel measurable. A measurable mixed strategy of Player 1 is a measurable $p : X \to [0,1]$; that of Player 2 consists of measurable $q_i : X \times [0,1] \to [0,1]$. Each value is the probability of action 1. Define

$$Q_i(x) = q_i(T_i x, t_{e,i}(x)), \qquad D(x) = s(x)\left(\sum_{i=1}^{3} Q_i(x) - \frac{3}{2}\right).$$

The maximal ex-ante unilateral gains R_1, R_2 are

$$R_1 = \frac{1}{2}\mathbb{E}_m\left[|D(x)|\,|p(x) - \mathbf{1}_{\{D(x)>0\}}|\right], \tag{7}$$

$$R_2 = \frac{1}{6}\sum_{i=1}^{3}\mathbb{E}_m \int_0^1 |p(y) - t|\,|q_i(y,t) - \mathbf{1}_{\{p(y)>t\}}|\,dt. \tag{8}$$

Indeed, integrating the sampled index in phase G gives Player 1's payoff difference $D/2$. Phase R gives Player 2's payoff difference $(p(y) - t)/2$ at each signal. Choosing the action with positive payoff difference is a measurable deviation, and pointwise maximization gives (7) and (8). A zero difference causes no difficulty: either action is optimal there.

Lemma 3. *The game (6) has no measurable mixed 2^{-40}-Nash equilibrium.*

Proof. Suppose $R_1, R_2 \le \varepsilon$ for some $0 < \varepsilon \le 1$. Put

$$h_i(x) = \mathbf{1}_{\{p(T_i x) > t_{e,i}(x)\}}, \qquad E(x) = \sum_{i=1}^{3} |Q_i(x) - h_i(x)|.$$

For fixed y, the interval $\{t \in [0,1] : |p(y) - t| < \eta\}$ has length at most 2η. Outside that interval, the unweighted relay error is bounded by its weighted error divided by η. Since $(T_i x, t_{e,i}(x))$ has law $m \otimes \mathrm{Leb}$, (8) gives

$$\mathbb{E}_m E \le 6\eta + \frac{6\varepsilon}{\eta}.$$

Take $\eta = \sqrt{\varepsilon}$. Markov's inequality then gives

$$\mathbb{E}_m E \le 12\sqrt{\varepsilon}, \qquad m\{E > 1/4\} \le 48\sqrt{\varepsilon}. \tag{9}$$

On $\{E \le 1/4\}$, $\sum Q_i$ is within $1/4$ of the integer $\sum h_i$. Its distance from $3/2$ is at least $1/4$, so $|D| \ge 1/4$ on this set. Let $c(x) = \mathbf{1}_{\{p(x)>1/2\}}$. Rounding to the closest endpoint gives $|p-c| \le |p-a|$ for every $a \in \{0,1\}$, as well as $|p-c| \le 1/2$. Therefore

$$\begin{aligned} \alpha := \mathbb{E}_m|p-c| &\le \frac{1}{2} m\{E > 1/4\} + 4\mathbb{E}_m[|D|\,|p - \mathbf{1}_{\{D>0\}}|] \\ &\le 24\sqrt{\varepsilon} + 8\varepsilon. \end{aligned} \tag{10}$$

Each root threshold $t_{e,i}$ is uniform and independent of $T_i x$. For fixed y, $\mathbf{1}_{\{p(y)>t\}}$ disagrees with $c(y)$ on a set of threshold values of length exactly $|p(y) - c(y)|$. Hence

$$m\{h_i(x) \ne c(T_i x)\} = \alpha. \tag{11}$$

Define

$$B(x) = \mathbf{1}_{\{s(x)(\sum_i c(T_i x) - 3/2) > 0\}}.$$

If $E \le 1/4$ and all $h_i = c(T_i x)$, then $\mathbf{1}_{\{D>0\}} = B$ and $|D| \ge 1/4$. If additionally $c \ne B$, then $|p - B| \ge 1/2$. The integrand in (7) before its factor $1/2$ is thus at least $1/8$. The union bound and (9)–(11) now imply

$$\begin{aligned} m\{c \ne B\} &\le 48\sqrt{\varepsilon} + 3\alpha + 16\varepsilon \\ &\le 120\sqrt{\varepsilon} + 40\varepsilon \le 160\sqrt{\varepsilon}. \end{aligned} \tag{12}$$

For $z = 2c - 1$, the identity $c = B$ is precisely

$$z(x) = s(x)\,\mathrm{Maj}(z(T_1 x), z(T_2 x), z(T_3 x)).$$

Taking $\varepsilon = 2^{-40}$ in (12) yields

$$\delta \le 160\, 2^{-20} = \frac{5}{32768} < \frac{1}{4096},$$

contradicting Lemma 2. □

4 Creating a common indifference row

Independently adjoin a fair flag $f \in \{o, d\}$ to Ω_0, creating the state space $\Omega = \Omega_0 \times \{o, d\}$. Player 1 observes its former signal together with this flag. Player 2 observes exactly (5) and does not observe the flag. Write Player 1's two mixed strategies as $p(x)$ on flag o and $r(x)$ on flag d. Player 2 still uses $q_i(y, t)$.

Define the new payoffs F_1, F_2 by

state	$F_1(\omega, a_1, a_2)$	$F_2(\omega, a_1, a_2)$
o, G, x, i	$a_1 s(x)(3a_2 - 3/2)$	0
o, R, y, i, t	0	$a_1 a_2$
d, G, x, i	a_1	0
d, R, y, i, t	a_1	$-t a_1 a_2$.

(13)

In every state,

$$F_j(\omega, 0, 0) = F_j(\omega, 0, 1) = 0, \qquad j = 1, 2. \tag{14}$$

The bounds are $|F_1| \le 3/2$ and $|F_2| \le 1$.

Lemma 4. *The game (13) has no measurable mixed 2^{-42}-Nash equilibrium.*

Proof. Suppose a profile is a ρ-equilibrium. Player 1 may deviate only on flag d, choosing action 1 there. The gain is $(1/2)\mathbb{E}_m(1-r)$, so

$$\mathbb{E}_m(1-r) \le 2\rho. \tag{15}$$

Player 1 may instead change only its strategy on flag o. Each such gain is one half its gain in the base game (6). The projected profile (p,q) consequently has base-game regret $R_1 \le 2\rho$.

Let q' be any measurable deviation of Player 2. Write g for its gain in (6) and $\widehat{g}$ for its gain in (13). With $\mathbb{E}_{y,i,t}$ denoting expectation under m, uniform i, and Lebesgue t, these gains satisfy

$$\widehat{g} = \frac{1}{4}\mathbb{E}_{y,i,t}(q'_i - q_i)(p(y) - tr(y)), \qquad \frac{1}{2}g = \frac{1}{4}\mathbb{E}_{y,i,t}(q'_i - q_i)(p(y) - t).$$

Since $|q'_i - q_i| \le 1$ and $0 \le t \le 1$, (15) gives

$$\left|\widehat{g} - \frac{1}{2}g\right| \le \frac{1}{4}\mathbb{E}_m(1-r) \le \frac{\rho}{2}.$$

The equilibrium bound $\widehat{g} \le \rho$ implies $g \le 3\rho$. Thus $R_2 \le 3\rho$, and (p,q) is in particular a 4ρ-equilibrium of the base game. At $\rho = 2^{-42}$ this contradicts Lemma 3. □

5 Exact stopping-game realization

Work on the state space Ω of (13) with its Borel sigma field and the specified common prior. Let A be generated by Player 1's signal, comprising the flag and its tree signal. Let B be generated by Player 2's signal (5). Set

$$\mathcal{F}_{1,0} = A, \quad \mathcal{F}_{1,1} = A \vee B, \qquad \mathcal{F}_{2,0} = \{\varnothing, \Omega\}, \quad \mathcal{F}_{2,1} = B. \tag{16}$$

These are increasing filtrations, and $\mathcal{F}_{2,t} \subseteq \mathcal{F}_{1,t}$ for both dates. If later dates are desired, keep the filtrations constant; the counterexample itself uses $T = 1$.

Write $F(a,b)$ for the vector $(F_1(\omega,a,b), F_2(\omega,a,b))$. Assign the first-stopping payoff vectors $X_{J,t}$ and no-stop vector Y by

t	$X_{\{1\},t}$	$X_{\{2\},t}$	$X_{\{1,2\},t}$
0	$F(0,0) = 0$	$F(1,1)$	$F(0,1) = 0$
1	$F(1,0)$	$F(1,1)$	$F(1,1)$

$$Y = F(1,0). \tag{17}$$

For player i, take the ith component as $X_{i,J,t}$ or Y_i. Every entry is a bounded measurable state function.

To include public survival without introducing spurious dependence on another player's seed, first express a strategy as its prescriptions on the two possible decision dates. Its date-0 prescription is a function of its initial signal and private seed. There is only one surviving public history at date 1: both players continued at date 0. The prescription on that history is a function of its date-1 private information and seed; its own earlier action and private memory are also determined by these inputs. Extend this survivor-history prescription counterfactually to all states and seeds, including paths on which the game has already ended. Define the resulting potential stopping time τ_i as 0 if its date-0 prescription stops, otherwise 1 if the extended date-1 prescription stops, and otherwise ∞. This potential stopping time is adapted to the player's own filtration and seed, and

agrees with its actual play up to termination. Thus the potential stopping times induce exactly the same realized payoff. If strategies are given directly as the randomized stopping times in the problem statement, use those stopping times themselves.

For any such potential randomized stopping profile, define

$$a = \mathbf{1}_{\{\tau_1 > 0\}}, \qquad b = \mathbf{1}_{\{\tau_2 \leq 1\}}. \tag{18}$$

The realized stopping payoff is exactly $F(a, b)$, pointwise in both the state and the seeds. If $\tau_1 = 0$, Player 1 belongs to the date-0 stopper set, and every outcome pays zero, which is row $a = 0$ in (14). If $\tau_1 > 0$ and $\tau_2 = 0$, Player 2 alone stops first at date 0, giving $F(1, 1)$. If both stopping times exceed 0, a date-1 stop by Player 2, alone or tied, gives $F(1, 1)$; a date-1 stop by Player 1 alone gives $F(1, 0)$; and no stop gives $F(1, 0)$. These are all possible cases, including simultaneous stops at both dates.

For each fixed private seed, the event $\{\tau_1 = 0\}$ is $\mathcal{F}_{1,0}$-measurable, and $\{\tau_2 \leq 1\}$ is $\mathcal{F}_{2,1}$-measurable. Thus a depends only on Player 1's initial signal and its own seed; b depends only on Player 2's signal and its own seed. Integrating the seeds produces measurable mixed strategies of (13). Their actions are independent conditional on the state because the seeds are independent. This is exactly the Bayesian game's mixed extension, even though Player 1 can acquire additional information at date 1: that information can change only whether it stops at 1 or never stops, and the payoff assignment already gives $F(1, b)$ in either case.

The counterfactual extension also shows why public survival preserves this reduction. In any path where Player 1 stopped at date 0, its binary action is $a = 0$, so its payoff is independent of Player 2's counterfactual action. If Player 2 stopped at date 0 while Player 1 continued, then $a = b = 1$ already determines the payoff, regardless of Player 1's counterfactual date-1 prescription. Conditioning on survival changes beliefs, but its incentive effect is already present in the ex-ante payoff expression $F(a, b)$. There is no additional surviving public branch or omitted signaling action.

Conversely, every measurable mixed strategy of (13) has an admissible stopping implementation. Using independent uniform seeds, Player 1 stops at 0 exactly when its binary action is 0, otherwise never stopping. Player 2 stops at 1 exactly when its binary action is 1, otherwise never stopping. These are randomized stopping times for (16). The same implementation works for each unilateral Bayesian deviation, and (17) preserves its expected gain exactly. Therefore a stopping ρ-equilibrium would induce a Bayesian ρ-equilibrium in (13). Lemma 4 excludes this at $\rho = 2^{-42}$, proving Theorem 1.

6 Scope, measurability, and normalization

The obstruction concerns the ex-ante equilibrium notion in the target: every measurable unilateral deviation must have gain at most the fixed ε. It already excludes this integrated notion, and hence also any stronger equilibrium notion that implies it. It treats all admissible randomized stopping strategies.

The construction satisfies arbitrary bounded state-dependent stopping payoffs and contemporaneously nested private information. It does not assert the same counterexample under an additional assumption that all payoff variables are measurable with respect to the less-informed player's private filtration, or that both initial private sigma fields are trivial. Neither restriction is present in the specified target.

For payoff range $[0, 1]$, replace every outcome payoff vector V in the stopping game by

$$\frac{1}{3}V + (1/2, 1/2).$$

The stated bounds ensure that both components then lie in $[0, 1]$. This divides every gain by three and preserves all comparisons, so the normalized game has no $(2^{-42}/3)$-equilibrium. The zero row becomes a common constant row, which preserves the exact realization.

All conclusions remain valid if the observation sigma fields are completed with respect to the common prior. Measurable strategy functions then have versions measurable under the underlying standard Borel signal laws. Changes on null signal sets do not alter the payoff integrals or the relay estimates: every subtree map T_i preserves m, and every relay signal has the stated product distribution.

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

References

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